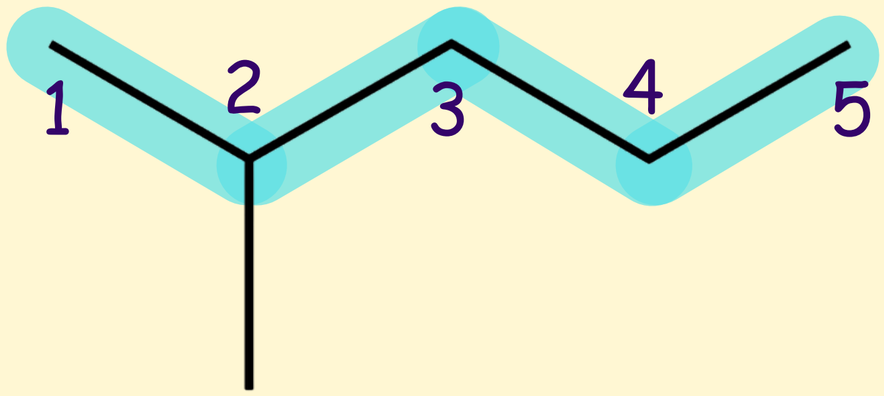
CARD A01 —— STRUCTURE ⟶ NAME

Level: FOUNDATION

Molecular Formula (Counted): C_6H_{14}

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = 5 C → pentane. Several possible 5-C paths exist, all carrying a methyl group.

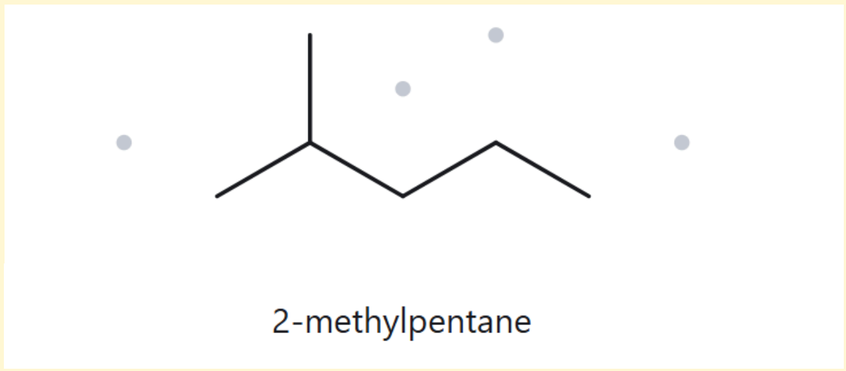
Numbering: left → right gives 2 and right → left gives 4. First point of difference: 4 > 2, so number from left to right.

Substituents: methyl on C2 → 2-methyl.

3. My Name

2-methylpentane

4. The alkane Check



Returned Name: 2-methylpentane

MATCH ✔

VERIFICATION SLIP

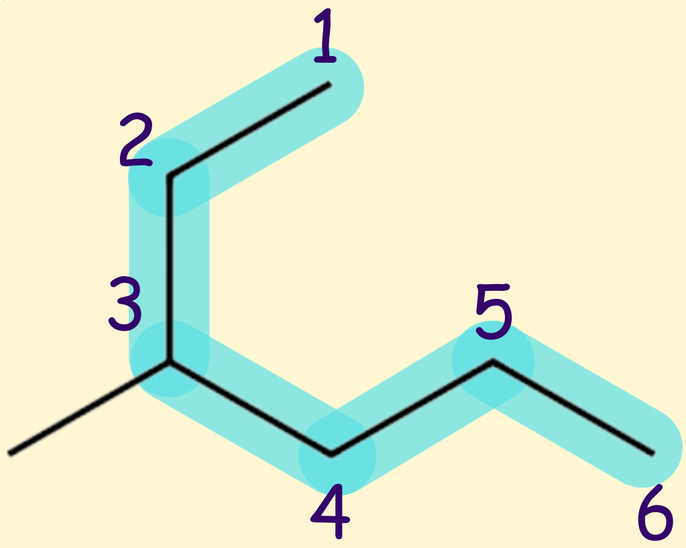
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A02 —— STRUCTURE ⟶ NAME

Level: FOUNDATION

Molecular Formula (Counted): C_7H_{16}

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = 6 C ➔ hexane. More than one possible 6-C paths exist, but all carry a methyl group.

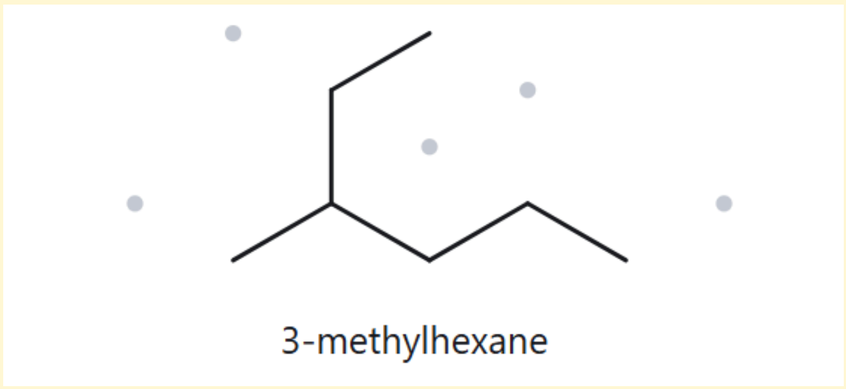
Numbering: top ➔ bottom right gives 3 and bottom left ➔ top gives 4. First point of difference: 4 > 3, so number from top to bottom right.

Substituents: methyl on C3 ➔ 3-methyl.

3. My Name

3-methylhexane

4. The alkane Check



Returned Name: 3-methylhexane

MATCH ✔

VERIFICATION SLIP

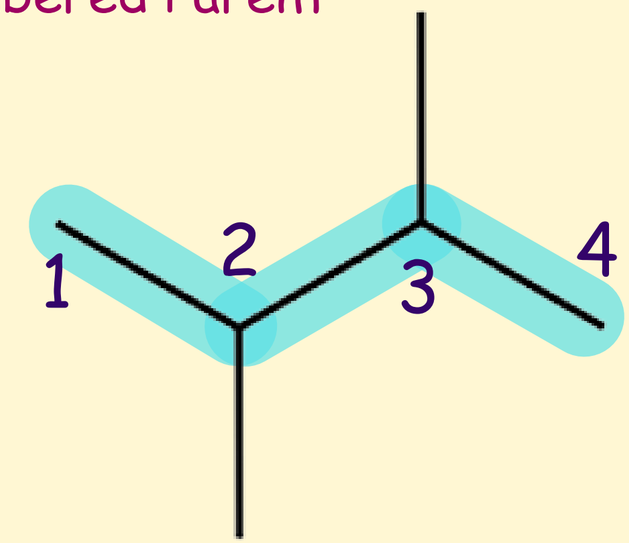
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A03 —— STRUCTURE ⟶ NAME

Level: FOUNDATION

Molecular Formula (Counted): C_6H_{14}

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = 4 C ➔ butane. More than one possible 4-C paths exist, but both carry a methyl group so they lead to the same name.

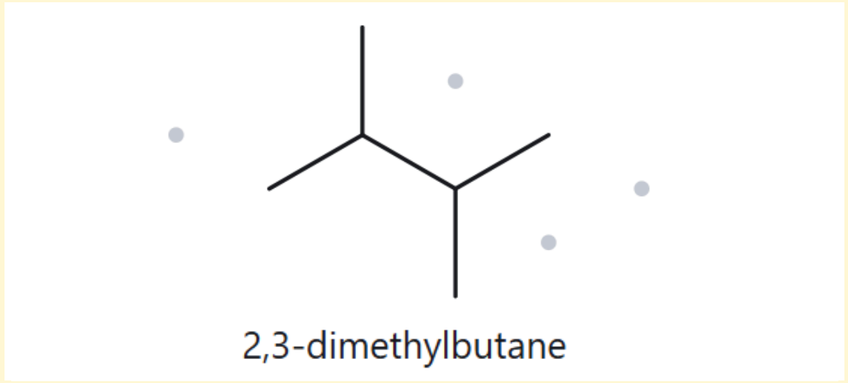
Numbering: left ➔ right gives 2,3 and right ➔ left also gives 2,3. Either way, both are numbered the same.

Substituents: methyl on C2 and C3➔ 2,3-dimethyl.

3. My Name

2,3-dimethylbutane

4. The alkane Check



Returned Name: 2,3-dimethylbutane

MATCH ✔️

VERIFICATION SLIP

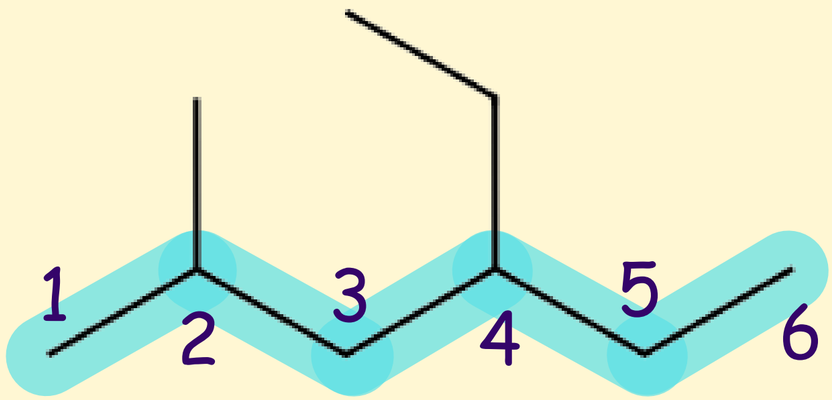
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A04 —— STRUCTURE ⟶ NAME

Level: DEVELOPING

Molecular Formula (Counted): C_9H_{20}

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = 6 C ➔ hexane. Several possible 6-C paths exist, but all carry the same methyl and ethyl group so they lead to the same name.

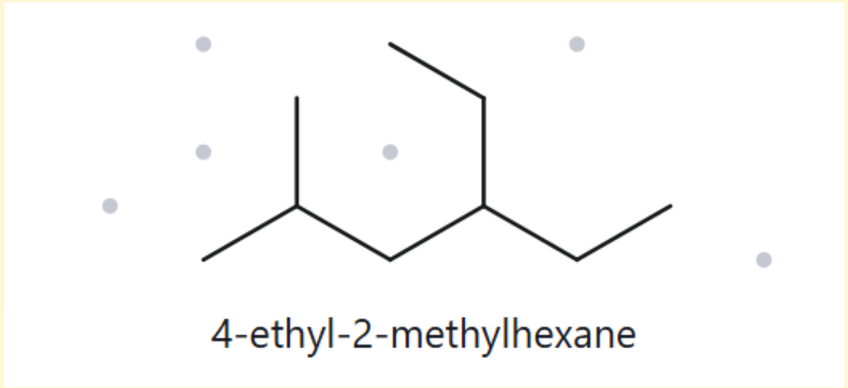
Numbering: left ➔ right gives 2,4 and right ➔ left gives 3,5. We choose 2,4 because it is the lower set at the first point of difference. Therefore, number from left to right.

Substituents: methyl on C2 and ethyl on C4 ➔ 4-ethyl-2-methyl.

3. My Name

4-ethyl-2-methylhexane

4. The alkane Check



Returned Name: 4-ethyl-2-methylhexane

MATCH ✔

VERIFICATION SLIP

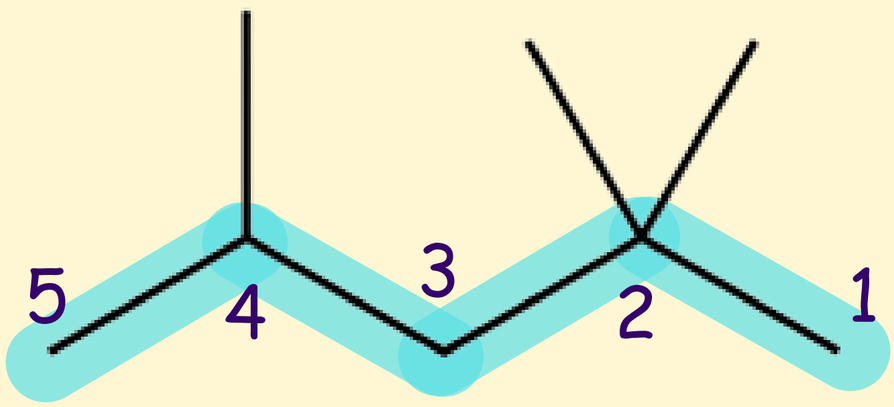
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A05 —— STRUCTURE ⟶ NAME

Level: DEVELOPING

Molecular Formula (Counted): C_8H_{18}

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = 5 C ➔ pentane. Several possible 5-C paths exist, but all carry 3 methyl groups.

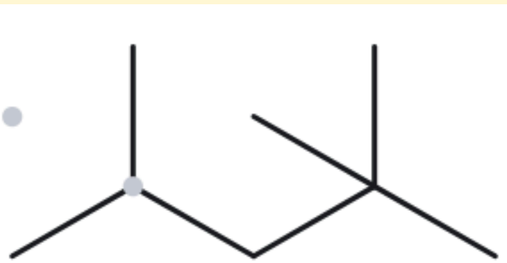
Numbering: left ➔ right gives 2,4,4 and right ➔ left gives 2,2,4. 4 > 2; 2,2,4 is lower at the first point of difference. Therefore, number from right to left.

Substituents: 2 methyl on C2 and one on C4 ➔ 2,2,4-trimethyl

3. My Name

2,2,4-trimethylpentane

4. The alkane Check



Returned Name: 2,2,4-trimethylpentane

MATCH ✔

VERIFICATION SLIP

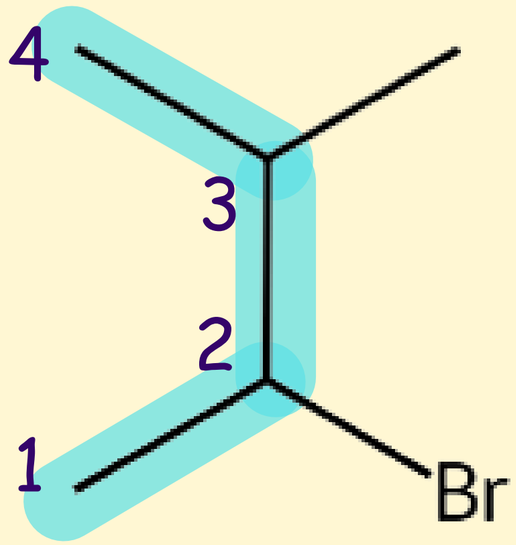
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A06 —— STRUCTURE ⟶ NAME

Level: DEVELOPING

Molecular Formula (Counted): $C_5H_{11}Br$

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = 4 C ➔ butane. More than one 4-C paths exist, but both carry a methyl and bromo group.

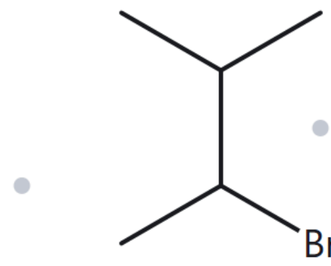
Numbering: bottom ➔ top gives 2,3 and top ➔ bottom also gives 2,3. Since both give the same set of numbers (2, 3), the tie is broken alphabetically: bromo comes before methyl, so bromo gets the lower number.Therefore, number from lower left-end to top.

Substituents: methyl on C3 and a bromo substituent on C2 ➔ 2-bromo-3-methyl

3. My Name

2-bromo-3-methylbutane

4. The alkane Check



2-bromo-3-methylbutane

Returned Name: 2-bromo-3-methylbutane

MATCH ✔

VERIFICATION SLIP

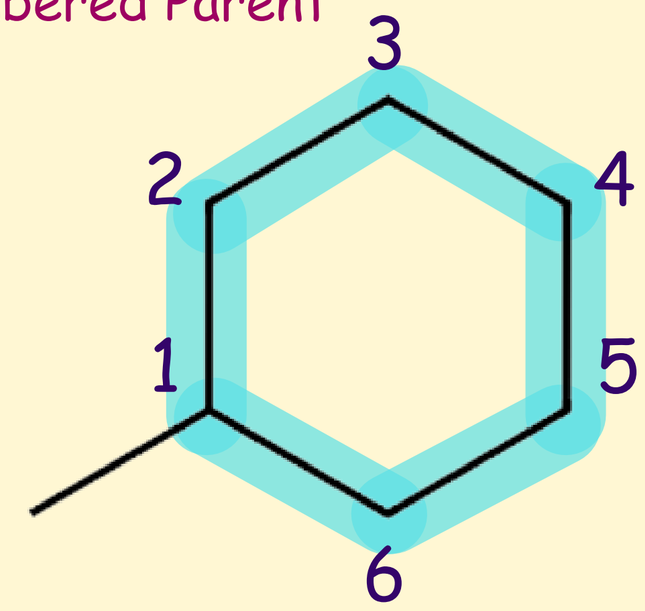
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A07 —— STRUCTURE ⟶ NAME

Level: DEVELOPING

Molecular Formula (Counted): C_7H_{14}

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = ring with 6 C ➔ cyclohexane. It carries a methyl group.

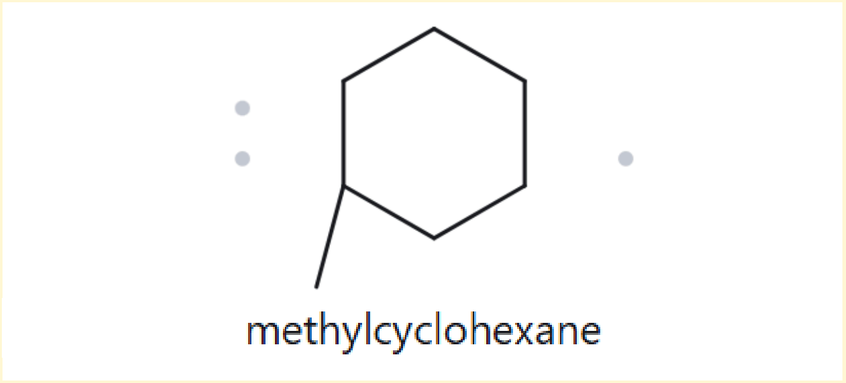
Numbering: There is no need to assign a number because there's only one substituent on the ring. That carbon is automatically C-1.

Substituents: methyl on C1 (but there's only one substituent on the ring; therefore, "1-" before methyl is unnecessary.) ➔ methyl

3. My Name

methylcyclohexane

4. The alkane Check



Returned Name: methylcyclohexane

MATCH ✔

VERIFICATION SLIP

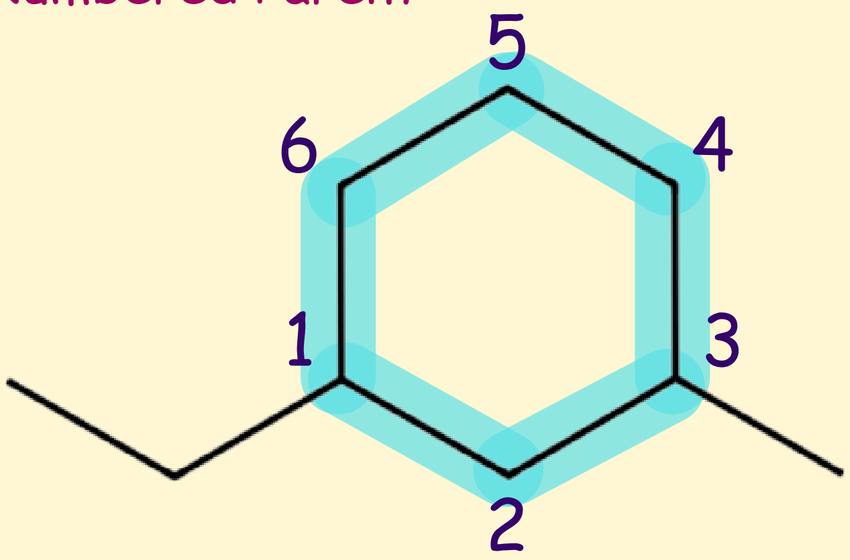
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A08 —— STRUCTURE ⟶ NAME

Level: CHALLENGE

Molecular Formula (Counted): C_9H_{18}

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = ring with 6 C ➔ cyclohexane. It carries a methyl and an ethyl group.

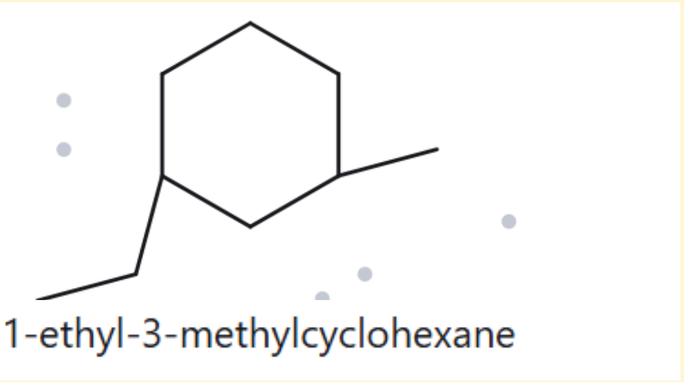
Numbering: Both directions give 1,3, but the substituent that comes first alphabetically (ethyl) gets C-1. Therefore, number from the ethyl group and counterclockwise.

Substituents: ethyl on C1 and methyl on C3 ➔ 1-ethyl-3-methyl

3. My Name

1-ethyl-3-methylcyclohexane

4. The alkane Check



Returned Name: 1-ethyl-3-methylcyclohexane

MATCH ✔

VERIFICATION SLIP

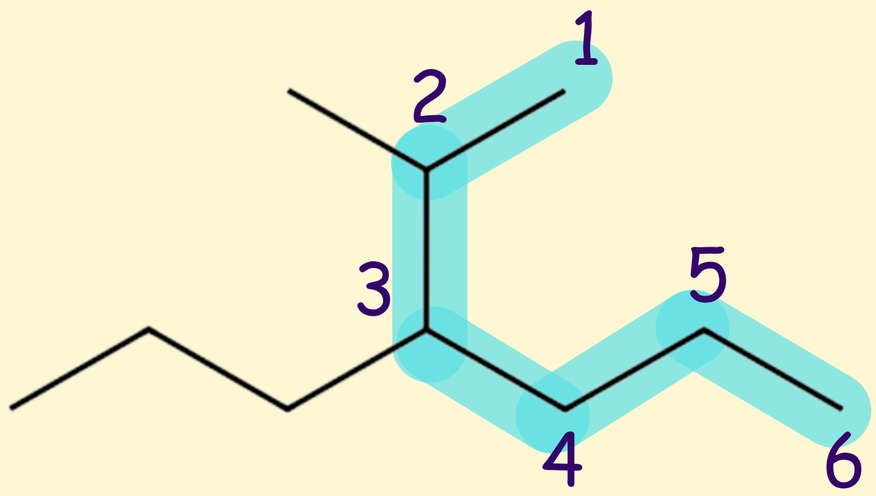
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A09 —— STRUCTURE ⟶ NAME

Level: CHALLENGE

Molecular Formula (Counted): $C_{10}H_{22}$

1. Numbered Parent



2. Numbered Trail

Parent: longest chain = 6 C ➔ hexane. It carries a methyl and propyl group.

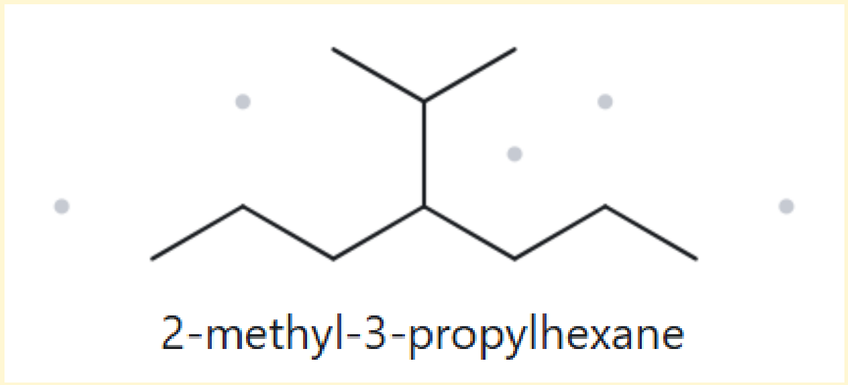
Numbering: top ➔ lower-end right gives 2,3 and top ➔ lower-end left also gives 2,3. Therefore, we can number from either side and it will still be the same.

Substituents: methyl on C2 and propyl on C3 ➔ 2-methyl-3-propyl

3. My Name

2-methyl-3-propylhexane

4. The alkane Check



Returned Name: 2-methyl-3-propylhexane

MATCH ✔️

VERIFICATION SLIP

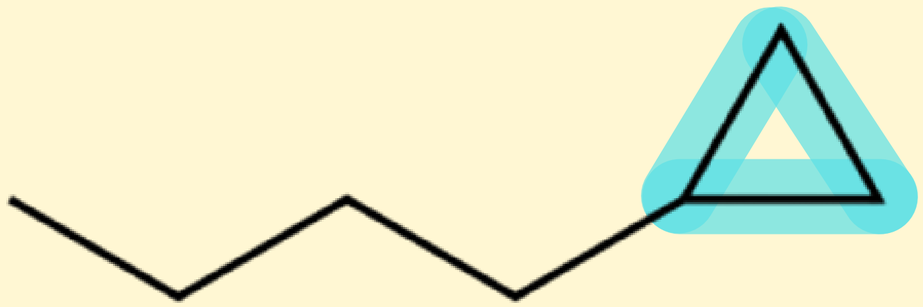
☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD A10 —— STRUCTURE ⟶ NAME

Level: CHALLENGE

Molecular Formula (Counted): C_7H_{14}

1. Numbered Parent



2. Numbered Trail

Parent: 3-carbon ring ➔ cyclopropane. It carries a 4-carbon chain substituent. The longest-chain rule does not override the ring-parent. Therefore, the ring is the parent.

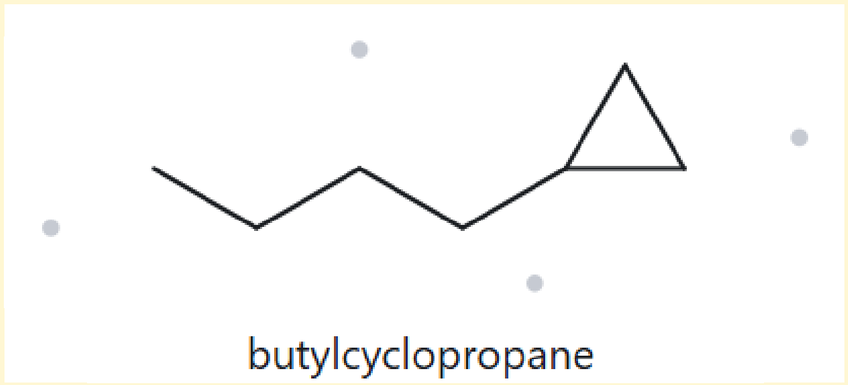
Numbering: The carbon of the cyclopropane ring where the butyl chain attaches is considered C1. However, because there's only one substituent on the ring, there is no need to include the locant in the name.

Substituents: butyl (A locant is unnecessary because there is only one substituent on the cyclopropane ring.) ➔ butyl

3. My Name

butylcyclopropane

4. The alkane Check



Returned Name: butylcyclopropane

MATCH ✔

VERIFICATION SLIP

☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted

CARD B01 —— STRUCTURE ⟶ NAME

Level: FOUNDATION

Given Name: 2-methylpropane

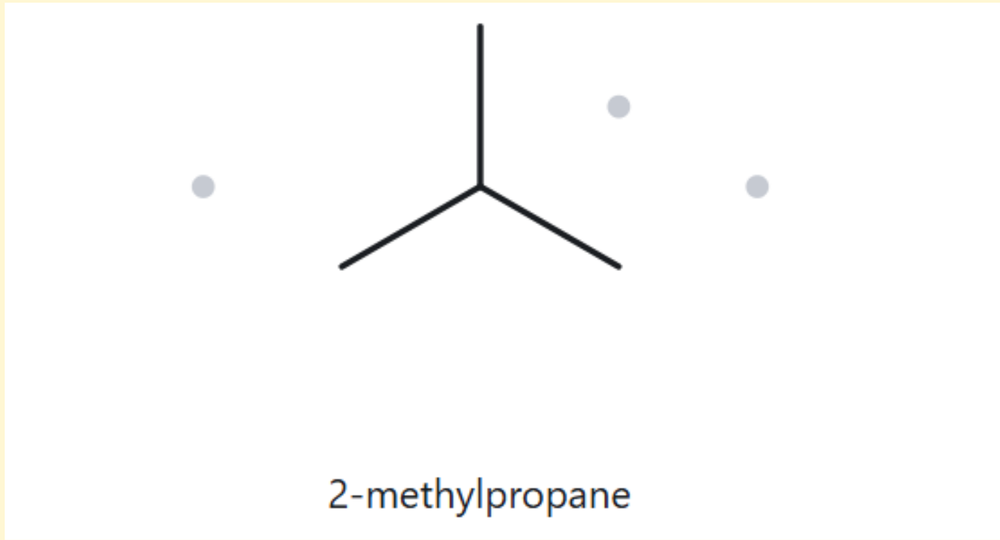
1. Name Decoded

Parent: propane → a 3-carbon chain.

Substituents: methyl on C2.

Expected Formula: 3 + 1 = 4 C → C_4H_{10}

2. Bond-line Structure (The Alkane)



2-methylpropane

3. Formula Check

Total Carbon atoms: 3 + 1 = 4

General Formula: C_nH_{2n+2}

Expected Formula: C_4H_{10}

Counted from the drawing: C_4H_{10}

4. Back Check

Returned by The Alkane: 2-methylpropane

✔ SAME AS THE GIVEN NAME

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B02 —— STRUCTURE ⟶ NAME

Level: FOUNDATION

Given Name: 3-ethylpentane

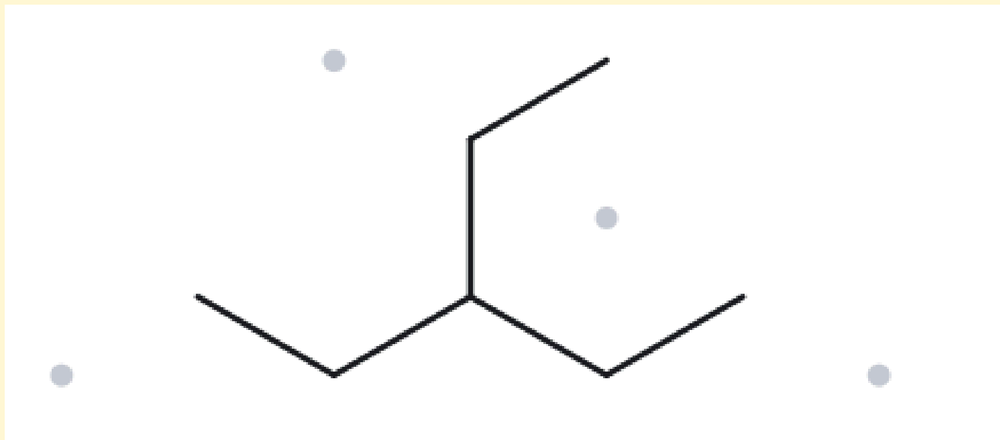
1. Name Decoded

Parent: pentane → a 5-carbon chain.

Substituents: ethyl on C3.

Expected Formula: 5 + 2 = 7 C → C_7H_{16}

2. Bond-line Structure (The Alkane)



3-ethylpentane

3. Formula Check

Total Carbon atoms: 5 + 2 = 7

General Formula: C_nH_{2n+2}

Expected Formula: C_7H_{16}

Counted from the drawing: C_7H_{16}

4. Back Check

Returned by The Alkane: 3-ethylpentane

✔ SAME AS THE GIVEN NAME

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B03 —— STRUCTURE ⟶ NAME

Level: FOUNDATION

Given Name: 2,2-dimethylbutane

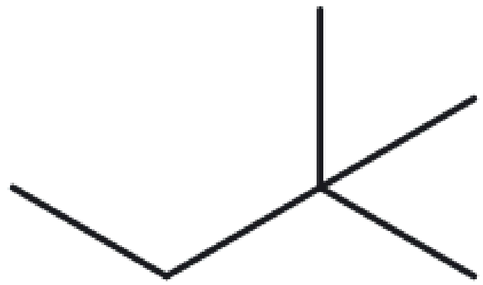
1. Name Decoded

Parent: butane → a 4-carbon chain.

Substituents: two methyl groups on C2.

Expected Formula: 4 + 2 = 6 C → C_4H_{10}

2. Bond-line Structure (The Alkane)



2,2-dimethylbutane

3. Formula Check

Total Carbon atoms: 4 + 2 = 6

General Formula: C_nH_{2n+2}

Expected Formula: C_6H_{14}

Counted from the drawing: C_6H_{14}

4. Back Check

Returned by The Alkane: 2,2-dimethylbutane

✔ SAME AS THE GIVEN NAME

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B04 —— STRUCTURE ⟶ NAME

Level: DEVELOPING

Given Name: 3-ethyl-2-methylpentane

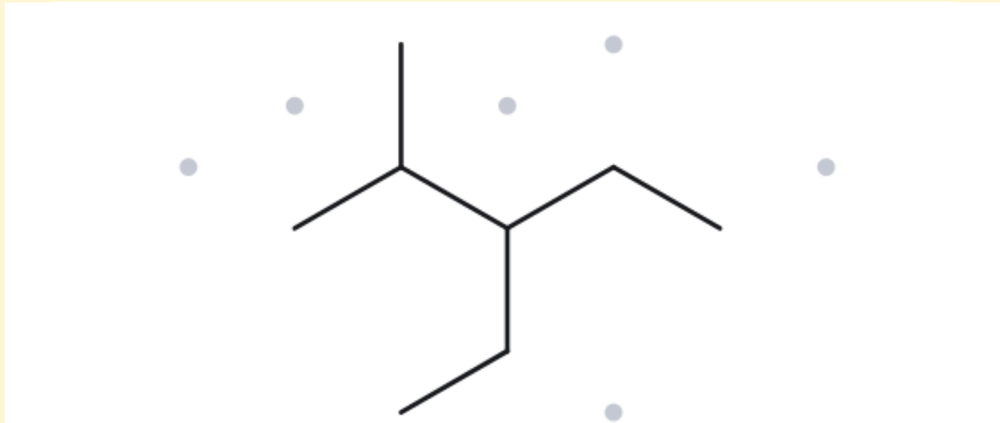
1. Name Decoded

Parent: pentane → a 5-carbon chain.

Substituents: an ethyl on C3 and methyl on C2.

Expected Formula: 5 + 2 + 1 = 8 C → C_8H_{18}

2. Bond-line Structure (The Alkane)



3-ethyl-2-methylpentane

3. Formula Check

Total Carbon atoms: 5 + 2 + 1 = 8

General Formula: C_nH_{2n+2}

Expected Formula: C_8H_{18}

Counted from the drawing: C_8H_{18}

4. Back Check

Returned by The Alkane: 3-ethyl-2-methylpentane

✔ SAME AS THE GIVEN NAME

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B05 —— STRUCTURE ⟶ NAME

Level: DEVELOPING

Given Name: 1-chloro-2-methylpropane

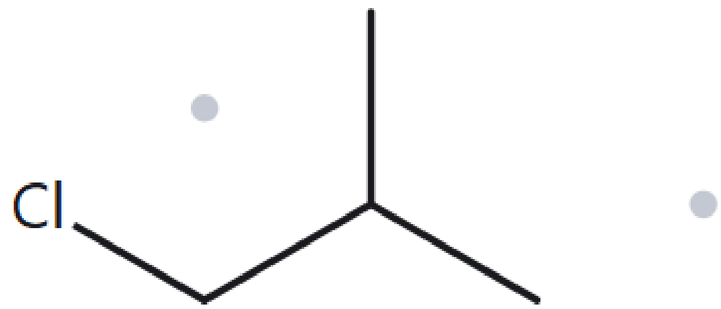
1. Name Decoded

Parent: propane → a 3-carbon chain.

Substituents: a chloro on C1 and methyl on C2.

Expected Formula: 3 + 1 = 4 C + 1 Cl; Cl replaces one H → C_4H_9Cl

2. Bond-line Structure (The Alkane)



1-chloro-2-methylpropane

3. Formula Check

Total Carbon atoms: 3 + 1 = 4

General Formula: C_nH_{2n+2}

Expected Formula: C_4H_9Cl ; minus one H for Cl

Counted from the drawing: C_4H_9Cl

4. Back Check

Returned by The Alkane: 1-chloro-2-methylpropane

✔ SAME AS THE GIVEN NAME

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B06 —— STRUCTURE ⟶ NAME

Level: DEVELOPING

Given Name: 1,2-dimethylcyclopentane

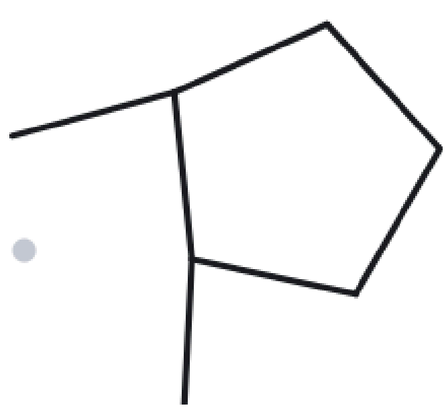
1. Name Decoded

Parent: cyclopentane ➔ a 5-carbon ring.

Substituents: two methyl groups on C1 and C2.

Expected Formula: 5 + 2 = 7 C ➔ C_7H_{14}

2. Bond-line Structure (The Alkane)



1,2-dimethylcyclopentane

3. Formula Check

Total Carbon atoms: 5 + 2 = 7

General Formula: C_nH_{2n} (for cycloalkane with one ring)

Expected Formula: C_7H_{14}

Counted from the drawing: C_7H_{14}

4. Back Check

Returned by The Alkane: 1,2-dimethylcyclopentane

✔ SAME AS THE GIVEN NAME

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B07 —— STRUCTURE ⟶ NAME

Level: DEVELOPING

Given Name: 4-ethyl-3,3-dimethylheptane

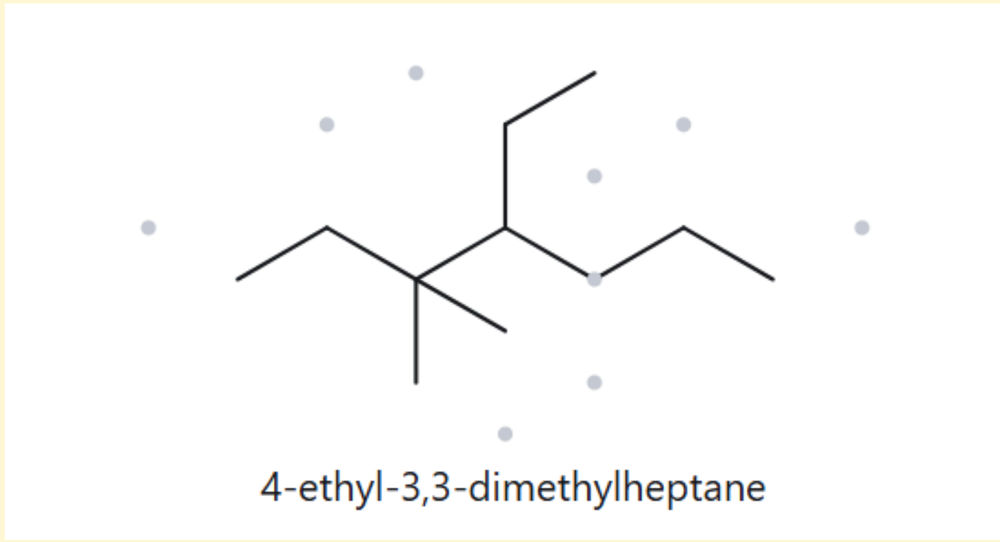
1. Name Decoded

Parent: heptane → a 7-carbon ring.

Substituents: an ethyl on C4 and two methyl on C3.

Expected Formula: 7 + 2 + 2(1) = 11 C → $C_{11}H_{24}$

2. Bond-line Structure (The Alkane)



4-ethyl-3,3-dimethylheptane

3. Formula Check

Total Carbon atoms: 7 + 2 + 1(2) = 11

General Formula: C_nH_{2n+2}

Expected Formula: $C_{11}H_{24}$

Counted from the drawing: $C_{11}H_{24}$

4. Back Check

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane

✔ SAME AS THE GIVEN NAME

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B08 —— STRUCTURE ⟶ NAME

Level: CHALLENGE

Given Name: tert-butylcyclohexane

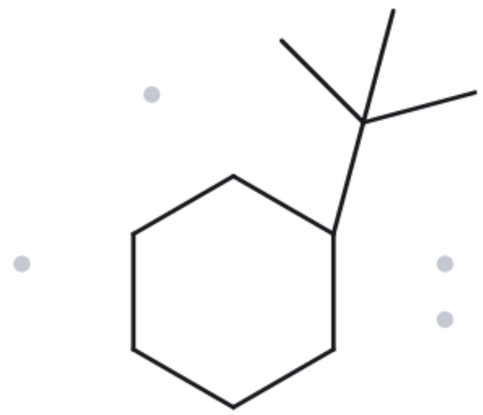
1. Name Decoded

Parent: cyclohexane → a 6-carbon ring.

Substituents: a tert-butyl on C1.

Expected Formula: 6 + 4 = 10 C → $C_{10}H_{20}$

2. Bond-line Structure (The Alkane)



(1,1-dimethylethyl)cyclohexane

3. Formula Check

Total Carbon atoms: 6 + 4 = 10

General Formula: C_nH_{2n} (for cycloalkane with one ring)

Expected Formula: $C_{10}H_{20}$

Counted from the drawing: $C_{10}H_{20}$

4. Back Check

Returned by The Alkane:

(1,1-dimethylethyl)cyclohexane

X NOT THE SAME AS GIVEN NAME

However, (1,1-dimethylethyl)cyclohexane and tert-butylcyclohexane are the same compound despite different names. tert-butyl is the common name for a substituent containing a central carbon attached to three methyl groups. Meanwhile, (1,1-dimethylethyl) is its systematic substituent name.

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B09 —— STRUCTURE ⟶ NAME

Level: CHALLENGE

Given Name: 3-ethyl-1,1-dimethylcyclohexane

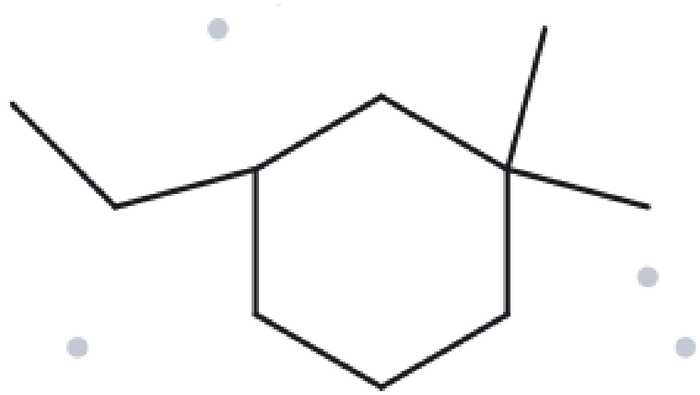
1. Name Decoded

Parent: cyclohexane → a 6-carbon ring.

Substituents: an ethyl on C3 and two methyl on C1.

Expected Formula: 6 + 2 + 1(2) = 10 C → $C_{10}H_{20}$

2. Bond-line Structure (The Alkane)



3-ethyl-1,1-dimethylcyclohexane

3. Formula Check

Total Carbon atoms: 6 + 2 + 1(2) = 10

General Formula: C_nH_{2n} (for cycloalkane with one ring)

Expected Formula: $C_{10}H_{20}$

Counted from the drawing: $C_{10}H_{20}$

4. Back Check

Returned by The Alkane: 3-ethyl-1,1-cyclohexane

✔ SAME AS THE GIVEN NAME

VERIFICATION SLIP

☑ Parent length matches ☑ Substituent positions match ☑ Formulas agrees ☑ Back-check recorded

CARD B10 —— STRUCTURE ⟶ NAME

Level: CHALLENGE

Given Name: 2-ethylbutane

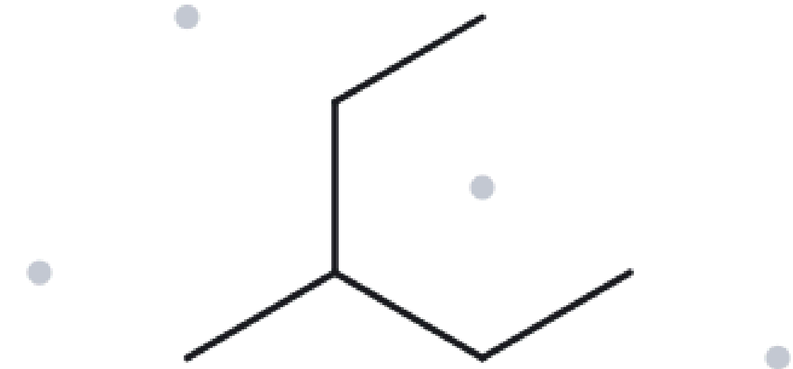
1. Name Decoded

Parent: butane → a 4-carbon chain.

Substituents: an ethyl on C2

Expected Formula: 4 + 2 = 6 C → C_6H_{14}

2. Bond-line Structure (The Alkane)



3-methylpentane

3. Formula Check

Total Carbon atoms: 4 + 2 = 6

General Formula: C_nH_{2n+2}

Expected Formula: C_6H_{14}

Counted from the drawing: C_6H_{14}

4. Back Check

Returned by The Alkane: 3-methylpentane

X NOT THE SAME AS GIVEN NAME

According to the IUPAC rules, we must always find the longest continuous carbon chain in the molecule. Tracing the longest chain actually gives us a parent chain including 5 carbons. Therefore, the correct name for the molecule would be 3-methylpentane.

VERIFICATION SLIP

☑ Longest chain/ring ☑ Lowest locants ☑ Alphabetical order ☑ Punctuation ☑ Formula counted